CS 6888: Program Analysis and its Applications
Assignment 0: Setup

Due: January 17 at 5:00 PM
Worth 5 points

1 Objective

This homework ensures that you can setup, execute, and submit an assignment.

2 Tasks

2.1 Install Docker

The tool/code part of each assignment will be using a docker container. You'll need to install `docker` and `docker-compose` which can both be obtained from your repos or together as part of "Docker Desktop" here: `https://www.docker.com/get-started`.

Note that you can only do this on a machine for which you have administrator access. For example you cannot do this on `portal.cs.virginia.edu`.

If you cannot get `docker` running on your machine, then you can use `apptainer` on `portal.cs.virginia.edu`. We provide instructions for that below.

2.2 Create a private fork of the class repo

Your code submission for each homework will be done through a git repo which we will checkout at the homework's due date. Note that your repo must be **private** and you will only have a single repo that contains all of the homeworks.

Github does not allow you to create a private fork of a public repo, so you will need to clone the `https://github.com/matthewbdwyer/cs6888-public` repo and then use `https://github.com/new/import` to create a new repo that you own and that is private. Share that private repo with `matthewbdwyer`.

2.3 Build a Local Image

There are two approaches to building an executable image for the homeworks you can use in the class. If you can use docker on your own machine then that is the simplest approach.

All of the commands mentioned below should be executed within the `cs6888-public` directory on your machine where you cloned your fork.

2.3.1 Local Docker Image

We will be using two docker images for the class. The first one is defined in the repo you just forked. Clone that and then run the following command to build and run it.

```
docker-compose up --build
```

Once it is running, you can use this command from another terminal to launch a shell inside the container. (Or you can just send the first command to the background.)

```
docker-compose exec main bash
```

Notes:

1. The build process may take 5-30 minutes depending on configuration and download speeds.

2. You will need $\sim$ 3GB free drive space and $\sim$ 2GB RAM to do the build. (More RAM on an arm mac, but should still run on the 8GB models.)

2.3.2 Local Apptainer Image

If you cannot use your own machine with Docker for some reason, you can use the CS `portal` nodes. Those nodes come with an open-source (near) equivalent to docker called Apptainer.

First you should load the apptainer module

```
module load apptainer
```

You can do add this to your login shell profile if you want to avoid typing it repeatedly.

We provide a pre-build docker image on Docker hub and you can use that to create a local Apptainer image with the following commands:

```
apptainer build main.sif docker://matthewbdwyer/cs6888-public-main
```

You can run this image as follows:

```
apptainer run main.sif bash
```

This will give you a shell with a prompt like `Apptainer>`. Apptainer does not translate all of the information from the docker image correctly, so to make the homework scripting work within the running Apptainer shell you should execute the following commands:

```
PATH=/root/infer/bin:/root/cppcheck/build/bin:$PATH
ln -s $HOME/cs6888-public/files $HOME/files
```

The second command you just have to set up one time, but the first you will need to enter each time you launch the image.

2.4 Check cppcheck's version

To make sure everything is working, from the container run `cppcheck --version`.

2.5 Modify run_all.sh

For grading each assignment, your code will be run with `run_all.sh` from the respective hwX directory - it is expected to recreate everything done in the assignment. You should make your other scripts as needed, and then call them from `run_all.sh`.

You should be able to navigate to `/root/files/hw0` with `cd files/hw0` inside the container; this directory is mapped to `cs6888-public/files/hw0` outside of the container. Depending on your operating system, you may need to change permissions for files created inside the container to view/edit them from outside.

For HW0, just add `cppcheck --version` to it and then push your changes back to github.

2.6 Create a report

For each assignment, in addition to your code, you will turn in a report in ieee-conference format as a pdf to Canvas. You can find a template on Overleaf at `https://www.overleaf.com/latex/templates/ieee-conference-template/grfzhhncsfqn`. If you haven't used LaTeXbefore, you can find an introduction at `https://www.overleaf.com/learn/latex/Learn_LaTeX_in_30_minutes`.

For HW0, please just put

1. your computing id
2. your repo address (in a `\url{}` tag)
3. the cppcheck version from above (in a `\verb||` tag)

3 Submission

Submit through Canvas under "Starter Homework":

1. The report as a `pdf` file.
2. The URL of your fork.